Canadian Community Health Survey (CCHS)

Household weights documentation

Statistique Canada Statistics Canada

Canada

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1. Introduction

This document describes the weighting process used in the creation of the household weights for the Canadian Community Health Survey (CCHS). The process is very similar to the process for other data products but here the product represents households rather than individuals. In using this weight, users should note that the survey is designed to collect information from individuals and not households. Certain steps are taken in the design to ensure that the sample is representative of different demographic groups and this may affect how well the sample represents households of different compositions. Also, since the calibration for the household weight is done at the provincial level, estimates are considered reliable only at the national and provincial level.

These weights should only be used for household-level variables. Such variables are those where the person-level responses clearly represent the household.

Note that combining cycles of CCHS data from before and after the redesign (e.g. combining 2014 and 2015 household files), or comparing estimates from before and after the redesign is not recommended because of the numerous changes in content and methodology.

2. Weighting Overview

To ensure estimates produced from survey data are representative of the target population and not just the sample itself, users must incorporate the survey weights in their calculations. A survey weight is given to each responding household included in the final sample. Each weight corresponds to the number of households in the entire population that are represented by the household of the respondent.

The CCHS uses two sampling frames for its sample selection: an area frame of dwellings to sample the Canadian population aged 18 and over, and a list frame from Canada Child Tax Benefit (CCTB) records to sample the population aged 12-17. Household weights are produced for those sampled from the area frame. Although the persons selected within these households are restricted to 18 years of age or older, this sample of households is a representative sample of Canadian households. Children less than the age of 18 also live in these households.

3. Sample weighting

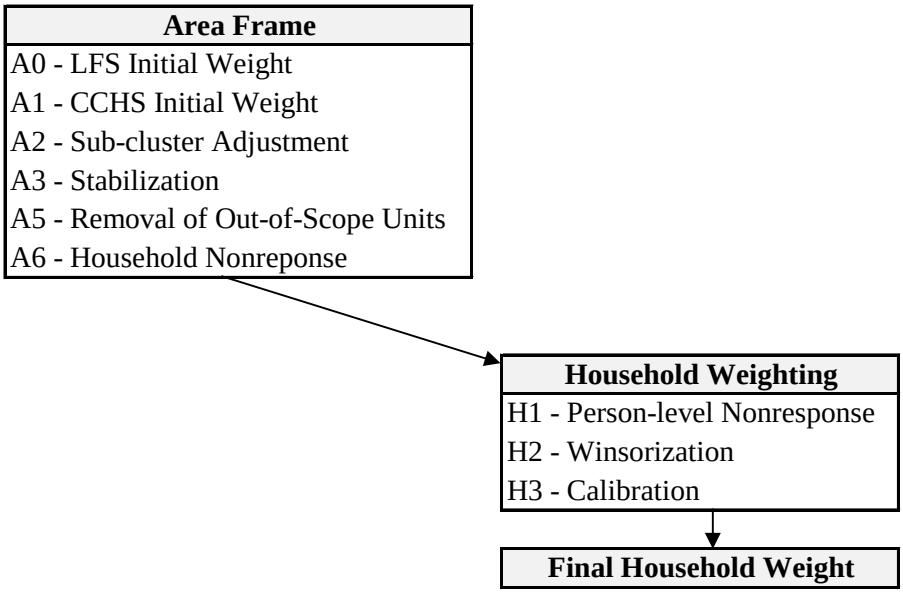
The household weighting steps up to (but excluding) person selection are the same as the steps from the CCHS person-level weighting process. Please refer to the CCHS User Guide for more information about these steps. The final three weighting steps deviate from the person-level strategy and are explained in sub-sections 3.1-3.3.

Following the redesign of the CCHS in 2015, due to operational constraints, estimates in the three territories will only be representative over two years. As such, respondents in the territories are omitted from the one-year files and will only be included on the two-year files. There are some

particularities associated with the weighting process for the territories, and this will be discussed in more detail in documentation of two-year CCHS files.

Diagram A presents an overview of the adjustments that are part of the household weighting strategy. Adjustments borrowed from the person-level process are enumerated from *A0* to *A6*. Please refer to the CCHS User Guide for more information about these steps. After step *A6* is where the household weighting process diverges from the person-level weighting process. These adjustments are enumerated from *H*1 to *H*3 for *H*ousehold.

Diagram A: Household Weighting Strategy Overview



3.1 Person-level Nonresponse (H1)

In some cases, interviewers obtain a household roster and select a member of the household but do not conduct an interview either because they cannot get in touch with the individual, or because that selected person refuses to be interviewed. Such individuals are defined as person nonrespondents, and an adjustment factor must be applied to the weights of person respondents to account for this nonresponse. In the case of the household weight, this person-level adjustment must still be applied since the selected person has not responded for the household.

Using a methodology similar to the one used in the treatment of household nonresponse (see the CCHS User Guide, Section 8.2 - A6), an adjustment is applied within response homogeneity groups. In this process, a scoring method is used to define a response probability based on characteristics available for both respondents and nonrespondents. Since it is the selected individual that did not respond for the household, the characteristics of this person are used in determining the response probability for the household.

The probabilities are grouped into response homogeneity groups within each province and the following adjustment factor is calculated within each group:

$$\frac{Sum\ of\ weights\ A6\ for\ all\ selected\ persons}{Sum\ of\ weights\ A6\ for\ all\ responding\ selected\ persons}$$

Weight A6 for responding persons is multiplied by the above adjustment factor to produce weight H1. Nonrespondents are dropped from the weighting process from this point onward.

3.2 Winsorization (H2)

Following the series of adjustments applied to the respondents, some units may come out with extreme weights compared to other units of the same domain of interest. These units could have a large impact on the variance. In order to prevent this, outlier weights are adjusted downward using a method referred to as “winsorization”.

3.3 Calibration (H3)

The last step in obtaining the final CCHS household weight is calibration. Calibration is done to ensure that the sum of the final weights corresponds to known household counts based on the Census. Monthly household counts are available and are defined at the province by household size level. The household sizes used are 1 person, 2 persons, and 3 or more persons.

Calibration is performed using Statistics Canada’s generalized estimation system, G-EST. This program adjusts the weights to ensure that the sum of the final household weights corresponds to the average of the monthly household counts at the province by household size level. At the same time, since data is collected throughout the year, the weights are adjusted to ensure that time periods are equally represented within the sample. For example, the households collected in one three-month collection period represent ¼ of the overall sum of weights.

The weight H2 is adjusted to obtain the final weight H3. Weight H3 corresponds to the ***final CCHS household weight*** that can be found on the household weight file with the variable name WTS_MHH.

3.4 Creation of the Share weight (WTS_SHH)

Along with the master file and PUMF, which contain all CCHS responding persons, a share file is created which contains only a portion (>90%) of the original CCHS responding persons. The individuals on this share file have agreed to share their data with certain partners. To compensate for the loss of some respondents from the file, the weights of these "sharers" must be adjusted by the factor:

$$\frac{Sum\ of\ weights\ H1\ for\ all\ responding\ persons}{Sum\ of\ weights\ H1\ for\ all\ responding\ persons\ agreeing\ to\ share\ their\ data}$$

Similar to the nonresponse adjustments, this factor is calculated within response homogeneity groups, where in this case, households with similar estimated propensity to share will be grouped together. Weight H1 of units accepting to share is multiplied by the above adjustment factor to produce a share weight. Winsorization and calibration, similar to adjustments H2 and H3, are applied to the share units. The final weight after these adjustments is called WTS_SHH.

3.5 The Food Security Module

The Food Security Module (FSC) is one of the few CCHS modules where it is appropriate to use the household weights. For 2017-2018, the FSC module is part of the theme content. For more information on theme content selection, please refer to the appendices of the User Guide.